

C++ Symbol and Type Recovery in Binary Ninja

REcon 2026

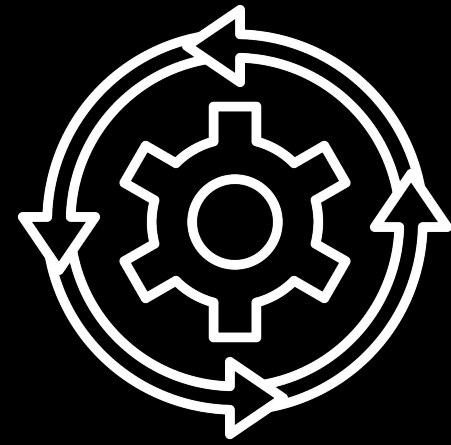
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./RE

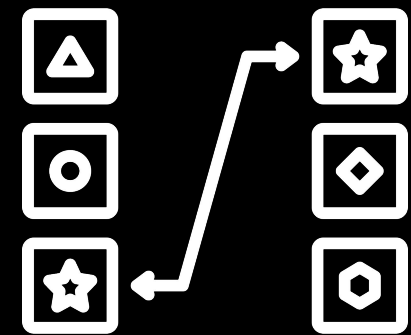
Follow Setup Steps in Manual

**WARNING: REAL MALWARE
USE A VM**

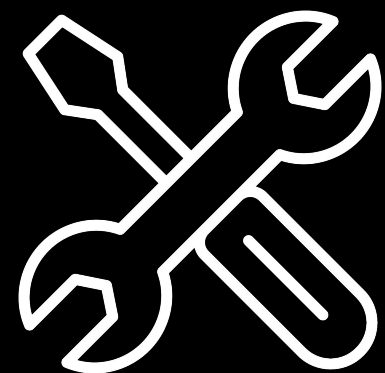
Advanced Binary Analysis



Data structure recovery via binary analysis

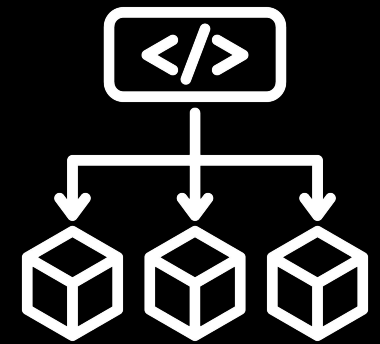


Type recovery via signature matching (Flirt, WARP)

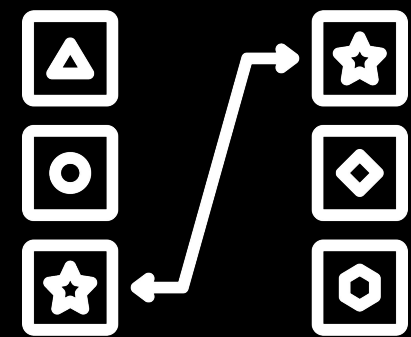


Diff tooling as a fallback

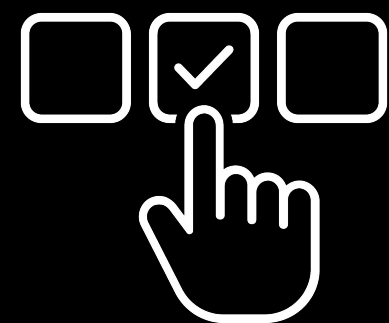
WARP



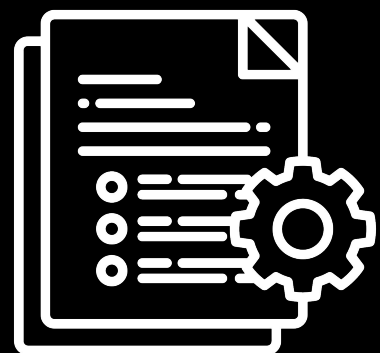
Function names, parameter names and types



Matching unique functions and applying information

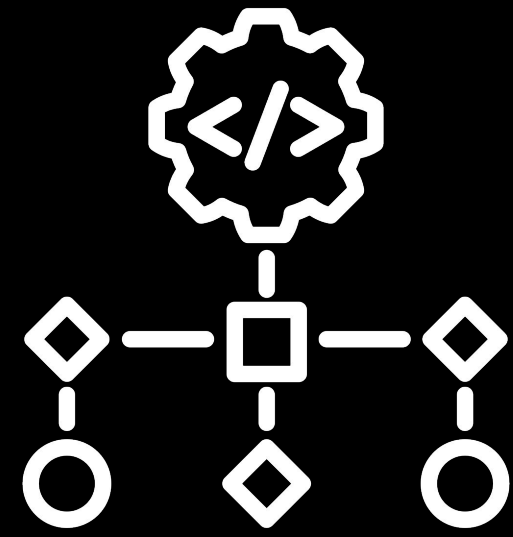


Allows selection of functions when failures occur

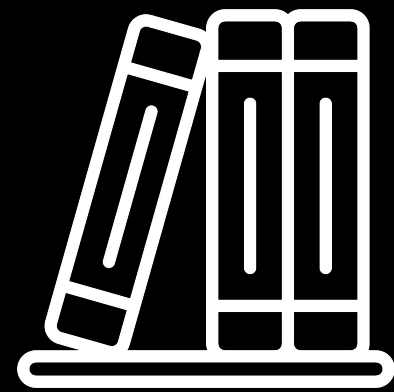


Instruction, basic blocks, registers, relocatable instrs

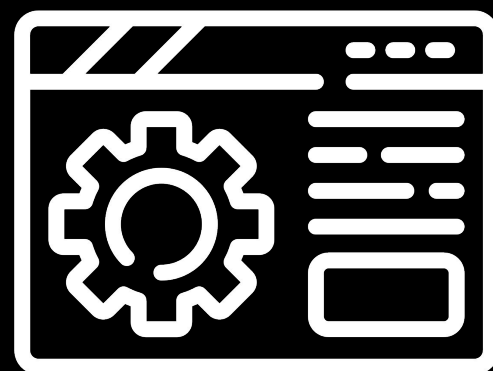
Why are symbols and type recovery needed?



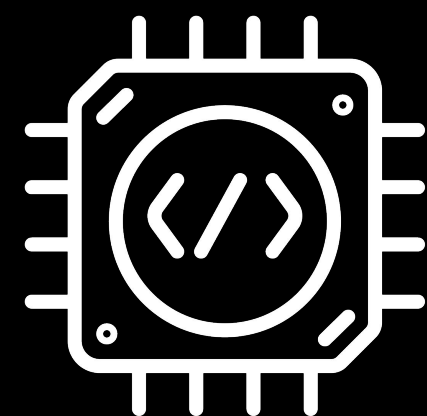
Data structures (arrays, structs, classes, lists)



Statically linked libraries

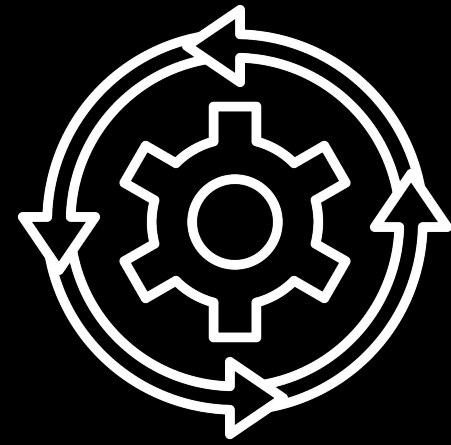


Managed binary analysis (Golang)

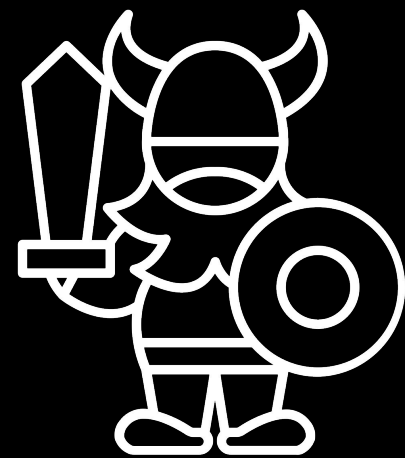


Unmanaged binary analysis (C++, Rust)

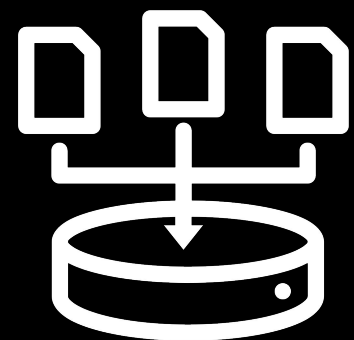
Debug Storage Formats



Program Database (PDB from MSVC)



Debugging With Attributed Record Formats (DWARF)



Can be applied to RE databases

References & Further Reading

- WARP. Retrieved October 31 2025, from <https://docs.binary.ninja/guide/warp.html>
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- National Security Agency. Ghidra BSim: Binary Similarity Matching Framework. Retrieved October 31, 2025, from <https://github.com/NationalSecurityAgency/ghidra/blob/master/GhidraDocs%2FGhidraClass%2FBSim%2FREADME.md>
- Microsoft. What's inside a PDB File? Retrieved October 31, 2025, from <https://devblogs.microsoft.com/cppblog/whats-inside-a-pdb-file/>
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Conclusion